JAVASCRIPT FUNCTION EXAMPLES

Example 1

07.example_1.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8" />
    <title>Function Example 1</title>
</head>
<body>
    <h1>Function Example 1</h1>
    <p>Please open the Console Window to see the program output.</p>

    <script src="07.example_1.js"></script>
</body>
</html>
```

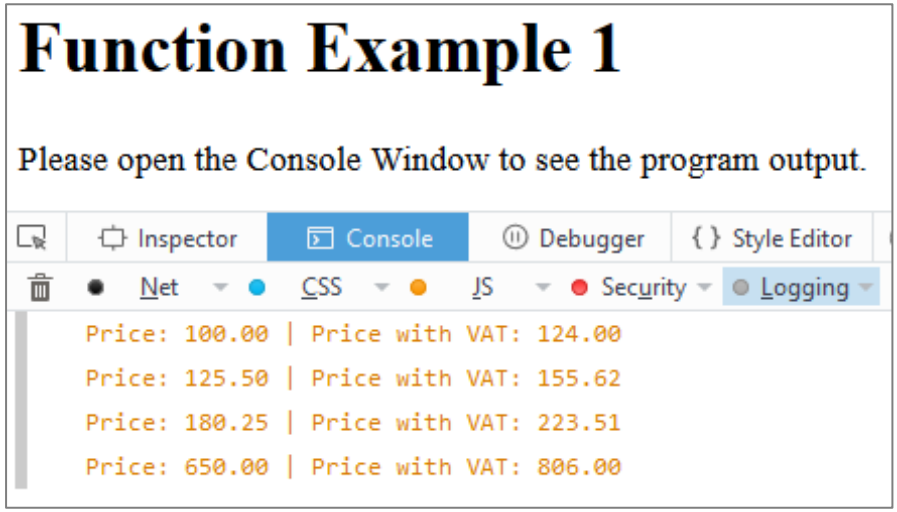
07.example_1.js

```
// Definition of the function displayPriceWithVAT
// The function is executed only when it is explicitly called.
function displayPriceWithVAT(price) {
    var priceWithVAT =  price * 1.24;

    var outputText = "Price: " + price.toFixed(2) +
        " | Price with VAT: " + priceWithVAT.toFixed(2);

    console.log(outputText);
}

// Making calls to the function
displayPriceWithVAT(100.00);
displayPriceWithVAT(125.50);
displayPriceWithVAT(180.25);
displayPriceWithVAT(650.00);
```



Function Example 2

Temperature in Celsius degrees: 38 Convert to Fahrenheit

38 degrees of Celsius is equal to 100 degrees of Fahrenheit

07.example_2.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8" />
    <title>Function Example 2</title>
</head>
<body>
    <h1>Function Example 2</h1>

    <label>Temperature in Celsius degrees: </label>
    <input type="text" id="txtTemperature" />

    <input type="button" value="Convert to Fahrenheit"
            onclick="showFahrenheit();" />

    <p id="pOutput"></p>

    <script src="07.example_2.js"></script>
</body>
</html>
```

07.example_2.js

```
// Definitions of the functions celsiusToFahrenheit and showFahrenheit

function celsiusToFahrenheit(celsius) {
    var fahrenheit = (celsius * 9) / 5 + 32;
    return fahrenheit;
}

function showFahrenheit() {
    var temperature = Number(document.getElementById("txtTemperature").value);
    var fahrenheitDegrees = celsiusToFahrenheit(temperature);

    var outputText = temperature.toFixed(0) + " degrees of Celsius is equal to " +
                     fahrenheitDegrees.toFixed(0) + " degrees of Fahrenheit";

    document.getElementById("pOutput").innerHTML = outputText;
}
```

Example 3

07.example_3.html

```
<!DOCTYPE html>
<html>
<head>
	<meta charset="UTF-8" />
	<title>Function Example 3</title>
	<script src="07.example_3_generic_functions.js"></script>
</head>
<body>
	<h1>Function Example 3</h1>

	<p id="pCurrentTime"></p>
	<p id="pOutputTax"></p>
	<p id="pOutputTemperature"></p>

	<script src="07.example_3.js"></script>
</body>
</html>
```

Function Example 3

Current time is 11:25

The tax is 43.20 euros

100 degrees of Fahrenheit is equal to 38 degrees of Celsius

07.example_3_generic_functions.js

```
// The functions below are independent of any particular web page.
// That is, they are 100 % reusable :-)

// Converts given Celsius degrees to Fahrenheit degrees
function celsiusToFahrenheit(celsius) {
	return (celsius * 9) / 5 + 32;
}

// Converts given Fahrenheit degrees to celsius degrees
function fahrenheitToCelsius(fahrenheit) {
	return (fahrenheit - 32) * 5 / 9;
}

// Returns current time as 'hh:mm'
function getCurrentTime() {
	var now = new Date();
	var hours = now.getHours();
	var minutes = now.getMinutes();

	return ("0" + hours).slice(-2) + ":" + ("0" + minutes).slice(-2);
}
```

07.example_3.js

```
//  Definition of the function calculateTax
function calculateTax(price, taxPercentage) {
   var tax = price * (taxPercentage / 100);
   return tax;
}

// Making a call to the function calculateTax
var tax = calculateTax(180.00, 24.00);

var outputText = "The tax is " + tax.toFixed(2) + " euros";
document.getElementById("pOutputTax").innerHTML = outputText;

//  Making a call to the function fahrenheitToCelsius
var celsiusDegrees = fahrenheitToCelsius(100);

outputText = "100 degrees of Fahrenheit is equal to " + celsiusDegrees.toFixed(0) + " degrees of Celsius";
document.getElementById("pOutputTemperature").innerHTML = outputText;

//  Making a call to the function getCurrentTime
document.getElementById("pCurrentTime").innerHTML = "Current time is " + getCurrentTime();
```